Vectored Thrust of a Ducted Fan with Jet Vanes by a Two-Way Coupled Rotating-Frame Vortex Lattice Method

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Control of a ducted-fan VTOL aircraft at hover rests on vanes immersed in the propulsor's own jet: at zero airspeed the propwash is the only dynamic pressure available to them. This paper presents a potential-flow method that resolves the complete propulsive wrench of such a system — rotor, duct, and exit vanes — in seconds per operating point. The rotor is solved as a vortex lattice in the blade-fixed rotating frame, with blades meshed on their CST camber surfaces wrapped on radius cylinders and a near wake of finite-core helical filaments — contracting and accelerating per momentum-theory kinematics, which keeps the hover limit well posed — whose radially-varying pitch is iterated to self-consistency with the solved loading, pulling the hover figure of merit into the physically admissible range where a straight-leg wake reads above the momentum-theory ideal. The duct is paneled into the same solve as constant-strength source rings, a two-way interaction converged with the blade circulation. The exit vanes cannot share the rotating frame; they are solved in the inertial frame, reading the rotor through a time-mean slipstream reconstructed from the converged radial loading by annular momentum theory, with a viscous sectional coupling that handles their separation-dominated hover aerodynamics. Vane loads close with the relation a stator row obeys: annulus-by-annulus angular-momentum bookkeeping, each strip's tangential force exactly the angular momentum it removes from its sector's mass flow, matched to blade-element aerodynamics at the passage-mean flow — so every force is bounded by its sector's momentum flux, and the recovered torque by what the shaft delivers, by construction. A block Gauss–Seidel alternation closes the loop between the two frames. On a reference cruciform at hover the coupled solve converges in nine outer passes, the vanes recovering ∼70% of the jet's angular-momentum flux as counter-torque while returning a net thrust credit through the pressure rise of swirl recovery, the system figure of merit holding at 0.48–0.50 across a fourfold rpm sweep. The method is implemented in the open-source C++23 toolkit Aeolion.

Nomenclature

c	section chord
c_l	section lift coefficient
F_t	vane-strip tangential force
FoM	hover figure of merit
$\dot{m}$	sector mass flow
M_x	rolling moment about the rotation axis
N_s	vane sectors per annulus

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p	body-axis roll rate
Q	shaft torque
q	dynamic pressure
r	radial station
R	rotor tip radius
r_c	wake filament core radius
s	swirl budget factor
T	thrust
u	axial jet velocity at the vane plane
v_i	annulus axial induced velocity
V_{ax}	axial onset velocity at the disk
V_{kin}	kinematic (rotational + freestream) local velocity
V_{rel}	local relative velocity
w_i	annulus swirl velocity
α_{eff}	effective section angle of attack
Γ	bound circulation
Δw	swirl removed by a vane strip's sector
δ	vane deflection angle
λ	inflow ratio
σ	source panel strength
Ω	rotor angular speed

Subscripts

i	radial strip index

I. Introduction

Tail-sitter VTOL aircraft of the ducted-fan class obtain control at hover from vanes placed in the duct's exit jet. Their sizing question is awkward for the established toolchain: at zero airspeed a conventional vortex lattice or panel method has no freestream dynamic pressure to put on a control surface, so vane authority, the thrust cost of vectoring, and the recovered swirl torque are outside such tools' model entirely, while RANS CFD of the full rotor–duct–vane system is too slow for the configuration loop — even the GPU-accelerated wall-modelled LES that has brought full-aircraft simulation to roughly one-day turnaround[1] is still a day per operating point. Classical VLM codes such as AVL[2] and OpenVSP's VSPAERO[3] treat the airframe with, at most, a prescribed actuator-disk propulsor; isolated-rotor methods such as XROTOR[4] carry no airframe and no vanes. Mid-fidelity solvers pairing potential boundary elements with Lagrangian wakes do reach interactional VTOL aerodynamics at orders-of-magnitude lower cost than CFD,[5] and potential-flow frameworks have recently been extended to powered jet wakes for aeropropulsive design;[6] what neither line closes is the loop this paper is about — a stator row standing in a ducted rotor's own hover jet, depleting the swirl that drives it. The components themselves are classical: shrouded propulsors and their static thrust augmentation are treated at length by Küchemann and Weber[7] and revisited experimentally for hovering micro air vehicles by Pereira;[8] vanes in a propulsive jet are among the oldest thrust-vectoring devices,[9] and duct lip shaping and exit control devices for hovering ducted-fan UAVs have been surveyed experimentally;[10] and swirl-recovery vanes have been wind-tunnel demonstrated on high-speed propellers.[11] What the established toolchain lacks is not the components but their coupled analysis at hover.

This paper describes a method that closes that gap at potential-flow-plus-sectional-viscous fidelity, resolving the coupled rotor–duct–vane system in seconds. Section II places the modelling choices against the cost–accuracy ladder the rotor literature has settled on, and states what each choice forfeits. Three ingredients are then presented in turn: the rotating-frame vortex lattice for the rotor (Sec. III), the two-way ducted solve and the inertial-frame vane solve with its reconstructed slipstream (Secs. IV, V), and the two-way rotor–vane alternation with its swirl momentum budget and cascade momentum closure (Sec. VI). Results on a reference cruciform configuration at hover follow in Sec. VII. The method is implemented in Aeolion, an open-source C++23 aerodynamics toolkit; the mathematical elements — horseshoe vortices and constant-strength source panels — are the textbook ones,[12] and the contribution is the coupling.

II. Fidelity Tier and Method Rationale

The choices described below are not free parameters, and this section states the warrant for each before the method itself is given. The cost–accuracy ladder for rotor aerodynamics is by now consistent across the review literature: blade-element momentum theory is cheapest and gives sound thrust trends but carries no tip-vortex or interaction physics; prescribed-wake filament methods follow; lifting-line and vortex-lattice blade models with prescribed or free wakes occupy the middle; hybrid vortex-lattice/vortex-particle and CFD-coupled schemes sit above, at correspondingly higher cost.[13,14] For ducted fans specifically, momentum plus blade-element hover models have been reported within roughly 15% of bench data[15] — adequate for sizing a thrust line, but silent about a vane row and structurally unable to carry the rotor–duct–vane interaction that sets control authority. At the far end, wall-modelled LES of a full aircraft has reached about one-day turnaround on GPU hardware.[1] The tier that fits the question is the one where potential boundary elements are paired with Lagrangian wakes: for unconventional VTOL configurations such methods reproduce CFD vertical force, pitching moment, and shaft power at orders-of-magnitude lower cost, and fail only where the flow is massively separated.[5]

That reported failure mode is precisely the hazard of the present configuration, and it is better named at the outset than buried in a limitations paragraph. At hover the rotor's own swirl parks the exit vanes at tens of degrees of local incidence before any control command, so the vanes are separation-dominated by construction — and vortex- and blade-element-type methods are known to underperform once viscous separation and post-stall nonlinearity dominate.[13] The design that follows is shaped by that fact: blades and vanes carry a sectional viscous coupling rather than a purely inviscid loading (Secs. III, V), and the default vane model is not a lifting surface at all but the angular-momentum closure a stator row obeys (Sec. A), adopted after the lifting-surface closure proved non-unique in exactly this regime.

Four further decisions are taken deliberately, each justified where it appears: a prescribed, load-consistent helical near wake with finite vortex cores instead of a force-free wake (Sec. III); a source-panel duct solved simultaneously with the blade circulation (Sec. IV); a slipstream reconstructed by annular momentum theory rather than sampled from the lattice's own wake field (Sec. V); and a partitioned two-frame iteration between the rotating and inertial domains rather than a monolithic unsteady solve (Sec. VI). Each buys speed at a stated cost, and those costs are collected in Sec. VI.

III. Rotor: Vortex Lattice in the Rotating Frame

The rotor is the same vortex lattice as a wing, posed in the blade-fixed rotating frame. Rotation is not special-cased in the solver: it enters as the body roll rate $p = \Omega$, whose kinematic velocity $\mathbf{V}_{kin} = \mathbf{V}_\infty - \boldsymbol{\omega} \times \mathbf{r}$ hands every blade panel its true Ωr tangential onset flow. The near-field Kutta–Joukowski force evaluation uses that same local velocity, so the body-axis load sums are dimensional propeller loads directly: thrust $T = -F_x$, shaft torque $Q = -M_x$, power $P = Q\,\Omega$.

Posing the blades in a rotating frame and holding the solution quasi-steady is the vortex-method counterpart of the multiple-reference-frame treatment that is standard for hovering rotors in CFD, where the governing equations are switched between a rotating and a stationary frame and the non-inertial terms enter at the source level.[16] Its price is the same in both settings — unsteady interaction effects are not carried — and it is acceptable here for a specific reason: hover of an axisymmetric rotor inside an axisymmetric duct is steady in the blade-fixed frame to begin with, and the quantities a configuration loop needs are cycle means. The blade model is a vortex lattice rather than a lifting line because resolving chordwise variation measurably improves hover loading and wake prediction over a one-dimensional spanwise representation while remaining far below particle or CFD cost;[13,14] it is a vortex lattice rather than blade-element momentum theory because blades, duct, and vanes must each see the others' induced field, which a per-annulus momentum balance cannot supply.

Each blade is meshed with one Weissinger row[17] per radial strip on the blade's CST camber surface,[18] with two geometric choices that proved load-bearing rather than cosmetic:

1. *Chords wrap on their radius cylinders* — the surface a blade element actually sweeps — rather than lying in a flat tangent plane. Near a small hub, where chord is comparable to radius, a tangent-plane chord subtends tens of degrees of azimuth, and the phantom incidence it picks up can overwhelm the real camber and twist loading.

2. *Trailing legs leave along the local relative wind* at their roots — the tangential sweep plus the axial convection — a linearized helix whose pitch is made *self-consistent with the solved loading*: the outer driver rebuilds the lattice each pass with the previous pass's banded induced velocity $v_i(r)$, obtained from the same annular momentum balance the slipstream reconstruction uses, so each leg convects at the axial inflow of its own radius, iterated to a fixed point. The classical momentum-theory hover inflow $\lambda \approx 0.07$ survives only as the seed pass's pitch and as a per-leg lower bound (a fraction of the blade's peak inflow) protecting unloaded stations from in-plane legs. Both directional ingredients are load-bearing: at hover the local wind is almost purely tangential, so trailing the wake axially leaves the legs near-perpendicular to the flow and the circulation solve diverges under refinement, while without the inflow floor a straight tangential leg lies in the rotor plane forever, crossing every radius outboard of its root.

A self-consistent *straight* leg is still not enough at hover: a straight filament cannot build the contracted helical vortex system that sits behind a real static rotor,[19] and it is that system's induction at the disk that carries most of the induced power — the straight-leg solve reads systematically optimistic there, with hover figures of merit *above* the momentum-theory ideal.[20] Each leg's first revolutions (two, at 12 segments per revolution) are therefore replaced by the helix a shed vortex particle traces in the rotating frame, discretized as a polyline: azimuth unwinds opposite the blade's sweep; the axial position convects at the leg's own $v_i(r)$, ramping linearly to the far-wake doubling $2v_i$ over a development length of $2R$; and the radius relaxes exponentially (length scale R) toward the far-wake area halving $r_0/\sqrt{2}$ that conserves mass through the accelerating jet. Beyond the helix the leg continues as the usual straight far tail along the direction at the helix end, already contracted and doubled, so the far field remains closed. Polyline segments induce through the same Biot–Savart kernel but with a finite Vatistas-style algebraic core,[21]

$$\mathbf{V} = \frac{\Gamma}{4\pi}\,\frac{h^2}{h^2 + r_c^2}\,(\cdots), \qquad r_c = 0.3\,\Delta r, \tag{1}$$

with h the perpendicular distance to the segment and Δr the shedding strip's radial width. The core is not cosmetic and it is not optional: a singular filament produces nonphysical induced velocities wherever the wake passes close to an evaluation point, so regularization is standard practice in filament methods.[13,22] Here the need is acute — successive helix passes bring neighboring blades' filaments within a discretization step of control points, and the raw $1/h$ kernel would make the solve hostage to exactly where a polyline vertex lands. A collinear polyline reproduces the straight-leg kernel identically, and an empty path *is* a straight leg, so fixed-wing cases are untouched.

The wake is prescribed rather than force-free by deliberate choice. Free-wake methods obtain the wake shape as part of the solution and are indispensable when wake deformation, rotor–rotor interaction, ground effect, or vortex-ring-state physics must emerge from the solution; prescribed wakes fix the shape from kinematic or empirical arguments and thereby avoid both the cost and the marching-stability difficulty of relaxing every wake node, which in hover is acute because many interacting vortex sheets must be marched together.[14,22] For repeated load evaluations under steady or mildly distorted conditions — which is exactly a hover design sweep — the prescribed choice is the recommended one, and semi-prescribed variants have been reported reproducing interaction-induced thrust and power changes within a few percent of experiment at a fraction of free-wake or particle cost.[23] The same partition by component appears in propeller–wing work, where a free wake on the propeller is paired with a prescribed wake on the downstream surface for speed.[24] What the present method adds is to make the prescribed helix a function of the solved loading, so that the wake shape is a fixed point of the solution rather than a fixed empirical prescription — addressing exactly the generalization across blade shapes and loadings that classical prescribed wakes are criticized for lacking.[14]

The lattice so equipped remains quasi-steady, its near-wake helix prescribed from momentum-theory kinematics in the lineage of the classical prescribed-wake hover analyses[25] rather than force-free, with no roll-up of the sheet into discrete tip vortices; it carries induced torque only and has no stall. The last two limitations are removed by a sectional viscous–inviscid coupling: per radial strip, the lattice provides α_{eff} from the local kinematic-plus-induced velocity, a 2-D section model provides $c_l(\alpha_{eff}, Re, Ma)$, the target circulation follows from the Kutta–Joukowski relation $|\Gamma| = \frac{1}{2} V_{rel}\, c\, c_l$, and the fixed point

$$R_i(\Gamma) = c_{l,i}^{VLM}(\Gamma) - c_{l,i}^{sect}\big(\alpha_{eff,i}(\Gamma), Re_i, Ma_i\big) = 0 \tag{2}$$

is converged with type-II Anderson acceleration[26,27] (67 plain-relaxation iterations reduced to 9 on the reference rotor), guarded by a kinematic-speed circulation cap that breaks the $\Gamma \to V_{rel} \to \Gamma$ runaway. The section model is an interface seam: an analytic viscous polar, or a transpiration-coupled integral boundary-layer solver (Thwaites,[28] Head,[29] Squire–Young[30]) whose displacement effect enters the 2-D section problem as transpiration boundary conditions — the viscous–inviscid interaction lineage whose standard exemplar is XFOIL.[31] The coupling caps the stalled hover root loading the inviscid lattice overpredicts and puts profile torque into the shaft-power sum.

IV. The Ducted Rotor

A boundary-element representation is the natural fidelity for the duct. Ducted fans are close to axisymmetric, and a panel formulation solves the inviscid external problem with unknowns on the surface alone rather than on a volume mesh, which is what makes shape studies and repeated design iterations affordable; unsteady panel analyses of ducted fans go back three decades.[32] The established weakness of this intermediate tier is equally clear: the lifting-line-plus-panel lineage that DFDC represents gives rapid and useful axial-flight thrust estimates, but tends to overpredict torque, and its agreement degrades where its axial-symmetry and incompressibility assumptions do.[33] Two of those weaknesses are addressed here directly — the duct is solved *simultaneously* with the blade circulation rather than applied as a prescribed correction, and profile torque comes from the sectional viscous coupling rather than from an inviscid lattice — while the third, viscous separation at the bore lip, stays outside an inviscid panel model and is the known reason such methods lose accuracy at high duct loading and in crosswind.[34]

The duct is meshed as a closed annular ring of constant-strength source panels (outer wall, bore wall, two caps) centered on the rotor plane, and passed into the coupled solve as its source system. The interaction is two-way: every coupling iteration re-solves the source strengths against the current blade circulation (the duct's no-through-flow condition sees the propwash; the source-only influence block is factored once), and the strip velocities include the duct's induced flow (the blades see the bore's constriction). One frame subtlety makes this legitimate: in the blade-fixed frame the duct rotates backwards, but a body of revolution about the rotation axis moves only tangentially to itself, so its boundary condition is unchanged (exactly, up to sector faceting).

The duct's own load is the surface-pressure integral $\mathbf{F} = \sum(\frac{1}{2}\rho|V|^2 - q_\infty)A\,\hat{\mathbf{n}}$, well behaved at hover where the gauge q_∞ is negligible — and this is where a shroud earns its keep:[7] the bore-lip suction ahead of the disk points upstream.

V. Exit Vanes and the Reconstructed Slipstream

The exit vanes are all-moving flat plates at the duct exit, spanning radially on their hinge axes and deflecting as rigid rotations about them. They cannot share the rotor's frame: a non-axisymmetric surface swept backwards through a rotating frame has a time-dependent boundary condition. The solve is therefore partitioned by frame — the vanes are meshed as ordinary lifting surfaces and solved in the inertial frame ($p = 0$), reading the rotor system through a prescribed external slipstream.

One meshing rule matters more than it looks: a vane whose span reaches the duct wall is recessed off it by 5% of span. A lifting line must not terminate on a neighboring source surface — the bound filament's tip endpoint would sit directly on the duct's bore, arbitrarily close to its source control points, where the filament's singular near field corrupts the shared no-through-flow boundary condition and, through the coupled source block, everything the duct touches. The physics says the same thing: a wall-attached vane sheds no free tip vortex at all — the wall carries the bound circulation onward as its image continuation — and the recession is the tip-clearance-scale stand-in for that image. A vane ending mid-jet keeps its stated tip and its real tip vortex.

That slipstream is *reconstructed*, not field-averaged. The quasi-steady rotor's prescribed straight trailing legs never wrap into the downstream helical wake cylinder, so their Biot–Savart mean carries swirl but almost no axial jet a few chords aft of the disk; vanes placed there would see no dynamic pressure. Instead, annular momentum theory[35] is applied to the loads the lattice actually converged: per annulus,

$$dT = 4\pi r\rho\,(V_{ax} + v_i)\,v_i\,dr \;\Rightarrow\; v_i, \qquad dQ = 4\pi r^3\rho\,(V_{ax} + v_i)\,w_i\,dr \;\Rightarrow\; w_i, \tag{3}$$

with dT, dQ summed over blades from the per-strip forces, the axial component developing toward the classical far-wake doubling over about a diameter, the swirl carried unchanged, and the duct's exact source induction added on top.

Annular momentum theory is a strong assumption set, and a bounded one. It supposes incompressible, axisymmetric, inviscid flow with the rotor replaced by an actuator disc, and it treats the annuli as independent; those assumptions are most defensible at high disc loading and where the aim is a low-cost reconstruction of the slipstream rather than blade-resolved near-wake detail[36, 37] — which is exactly the use made of it here, since the vanes need the mean axial and swirl profile that drives them, not the tip-vortex structure. The assumption carrying the largest known error is annulus independence: exact potential-flow solutions for the actuator disc show that annuli are in fact coupled through the pressure field, and that the axial velocity at the disc is not radially uniform.[36, 38] Both effects are partly mitigated in the present construction — the radial distribution is taken from the converged lattice loading rather than assumed uniform, and the duct's own source induction is superposed exactly — but what reaches the vane plane is still an axisymmetric, undeformed jet. Real slipstreams deform measurably before they meet a downstream surface, with root and tip vortex systems rolling up and reorganizing the local flow.[39] That deformation is outside this model, and the vane loads it returns are accordingly passage-mean loads.

The vanes receive the same viscous sectional coupling as the blades, with each trailing leg leaving along the *local mean flow* at its root — in the jet, helically, with the swirl. At hover the swirl alone puts the vanes at tens of degrees of local incidence before any command, so their hover aerodynamics are separation-dominated by nature: the regime in which vortex- and blade-element-type methods are least reliable[13] and in which mid-fidelity VTOL solvers are reported to break down.[5] The experimental swirl-recovery-vane literature reports the same failure concretely, with inboard vane sections stalling and limiting performance until their local pitch is raised.[40] A purely inviscid vane model is therefore not defensible at hover. The section model is assigned where each is honest: the budget-coupled solve uses the smooth analytic polar (nearly the whole span sits outside the boundary-layer solver's validity envelope in the hover jet, and the budget iteration needs the polar's smoothness to close), while the transpiration-coupled boundary-layer model is used where the flow is attached — vanes in brisk attached-incidence jets, and the blades' outboard strips at every condition. Three numerical safeguards proved necessary: the coupling residual is weighted by local dynamic pressure (a vane tip parked at 85° incidence in the jet's shear edge at 7% of the jet's q must not hold the solve hostage); a flat-plate deep-stall decay in the polar is what makes such strips satisfiable at all; and reversed-flow strips ($|\alpha_{eff}| > 75°$) are excluded from the convergence measure outright — strip theory has no lift-matching contract there, and such a strip both wedges the residual open and, when it runs away, inflates the reference dynamic pressure until the raw state reads converged. With the exclusion the hover vane solve converges where it previously hit the iteration cap.

The propulsive wrench is the sum of the two solves' near-field loads about the hub: rotor circulatory and profile forces, duct surface pressure, vane circulatory and profile forces. The vanes' drag debits the net thrust (though a de-swirling stator can return a net thrust credit through its pressure rise, Sec. VII); their side forces and M_y/M_z are the control authority — real at zero airspeed, the propwash being their only dynamic pressure — and even undeflected they recover part of the swirl as counter-torque.

VI. Two-Way Coupling and the Swirl Momentum Budget

Coupling a rotating component to a stationary one admits three broad treatments. A rotating-frame formulation switches the governing equations between frames and is efficient for steady and quasi-steady problems, at the price of neglecting unsteady interaction.[16] An inertial-frame unsteady solve with overset or embedded grids is the most faithful and by far the most expensive, and is the accepted reference when blade-resolved unsteady interaction is the object of study. Partitioned iterative coupling sits between them: each domain keeps its own solver, and interface data are exchanged until the pair is consistent, trading some stability margin for modelling flexibility and cost.[41] The present method is the third, with the rotating and inertial domains as its two partitions, and it takes the assumptions this class of coupling is normally granted — circumferential symmetry, a prescribed (here reconstructed) wake transfer between the domains, and tolerance of near-rotor discrepancies where the target is the first-order installation effect rather than blade-resolved unsteady loads. Azimuthal averaging in the vane-to-rotor direction, Eq. (4) below, is the concrete form circumferential symmetry takes here.

Partitioned as above, the scheme is one-way: the vanes read the jet, the rotor never feels the vanes, and nothing depletes the prescribed swirl — a high-solidity vane set can recover more torque than the jet carries.

The two-way scheme closes the loop as a block Gauss–Seidel alternation between the frames; the complete solution procedure is charted in Fig. 1. Two vane closures ride on the same alternation: the lifting-surface (lattice) closure described first, whose swirl exchange must be constrained by an explicit budget, and the cascade momentum closure described below — the default — which satisfies the budget by construction. Per outer pass, under the lattice closure:

1. *Rotor remesh and solve with the duct* (rotating frame). The blade lattice is rebuilt with the previous pass's banded inflow $v_i(r)$ setting the trailing-leg pitch (Sec. III), so wake pitch, vane feedback, and the swirl budget converge together in the same outer loop; the vanes enter through the external-velocity hook as the azimuthal mean of their induced field,

$$\bar{\mathbf{v}}_{vane}(P) = \frac{1}{K}\sum_{j=0}^{K-1} R_x(-\theta_j)\,\mathbf{v}_{vane}\big(R_x(\theta_j)P\big), \qquad \theta_j = 2\pi j/K. \tag{4}$$

   A blade sweeps past the static vane system once per revolution, and its quasi-steady condition sees the time mean of that encounter. Averaging is legitimate in *this* direction because the vane system's near field is fully represented by its explicit legs — it is exactly the operation that fails in the other direction.

2. *Slipstream reconstruction* from the fresh rotor loading, Eq. (3), with the swirl scaled by the budget factor $s \in [s_{min}, 1]$.

3. *The swirl budget, solved as a root problem.* The angular-momentum flux the jet delivers is the shaft torque, $\dot{L}_{jet} = Q = -M_x^{rotor}$; the vanes cannot remove more than arrives, yet an unconstrained prescribed-field vane solve can (nothing depletes the stated swirl). With the rotor state *frozen*, $M_x^{vane}(s)$ is a monotone, bracketed function of the swirl factor — each evaluation remeshes the vanes in the rescaled slipstream (their trailing legs follow the local mean flow) and re-solves them — so the budget condition

$$M_x^{vane}(s) = Q \tag{5}$$

   is solved directly by bracketed bisection on $s \in [s_{min}, 1]$, warm-started from the previous pass's factor, so a settled case costs a single evaluation. A slack budget ($M_x^{vane}(1) \leq Q$) accepts $s = 1$; a configuration over-recovering even at the floor keeps the floor, with the excess held in the convergence residual rather than declared away. Freezing the rotor inside the root find is load-bearing: driving s with a feedback law while the rotor re-solved was observed to fail both ways — a one-way multiplicative contraction ratchets past the fixed point onto the floor after a severe first-pass over-recovery and can never climb back, while a symmetric bidirectional law lets the co-evolving rotor–vane pair walk onto a runaway branch that "converges" with the rotor at several times its true loading.

4. *Rotor feedback.* Only after the budget is satisfied does the pass advance the vanes' circulation into the rotor's boundary condition.

5. *Relaxation and convergence.* The vane circulation fed back to the rotor is under-relaxed, deliberately gently: the swirl-rich hover jet parks vane strips in regimes where their inner solve plateaus, and an aggressive feedback amplifies that scatter. The outer residual is the relative change of the total wrench together with the vane-circulation change and any remaining budget excess.

Two further stabilizers earn their keep in the swirl-rich jet. The root find runs only through the first half of the outer pass budget: once the factor's increments fall inside the vane solve's own noise, re-bisecting each pass just remeshes jitter into the wrench, so later passes hold s and let the rest of the state settle. And the vane solve is warm-started from its previous circulation — continuation across evaluations and passes. This matters because the post-stall section response is multivalued: a deflected hover vane re-solved from scratch can land on either branch under tiny input changes, which shows up as a never-decaying pass-to-pass oscillation in the outer residual, while warm-started it follows one branch smoothly. Relatedly, an inner solve that exhausts its iteration budget in a limit cycle (deep post-stall strips do this) returns the cycle *mean* — the arithmetic mean over the second half of its iterates, evaluated consistently in one final sweep — rather than a random phase of the cycle, and reports itself unconverged so the residual keeps the cycle's mismatch.

Still outside the model, by construction: unsteady blade-passing loads (every exchange is a time mean) and the vanes' straight-line wakes. The lattice closure's budget is additionally a single scalar — radially resolved swirl depletion is exactly the band-by-band bookkeeping the cascade closure below supplies.

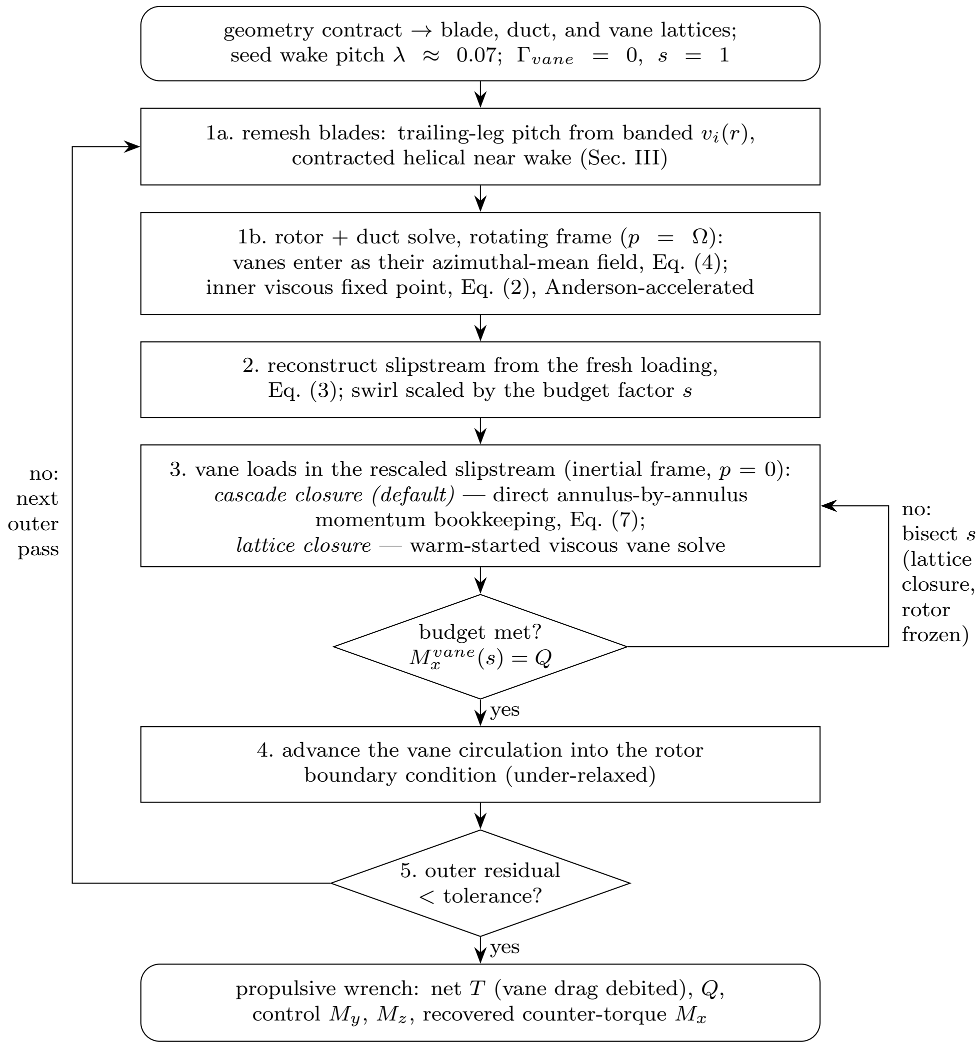

Figure 1. The coupled solution procedure; step numbers follow the enumeration in the text, with step 1 split into the blade remesh (1a) and the rotor + duct solve (1b). Under the default cascade closure (Sec. VI.A) the budget is satisfied by construction — the sampled swirl is normalized once to the jet's flux, s stays 1, and the root-solve loop is skipped. Under the lattice closure the inner loop on the right is the swirl budget solved as a bracketed root problem on $s \in [s_{min}, 1]$ with the rotor state frozen: a slack budget ($M_x^{vane}(1) \leq Q$) accepts $s = 1$, over-recovery at the floor keeps s_{min} with the excess held in the outer residual, and later outer passes hold the settled factor. The outer loop on the left is the block Gauss–Seidel alternation between the rotating and inertial frames.

A. The Cascade Momentum Closure

The machinery above still asks isolated-strip lifting-surface theory to carry vanes parked in tens of degrees of hover swirl, and the honest verdict from exercising it across an rpm–deflection matrix is that post-stall the model's answer is *not unique*: each strip's circulation is individually capped, but twenty capped strips can mutually inflate one another's local velocities, so the force runs away with no Γ ever exceeding its cap — a single vane at 8° was solved carrying more side force than the entire net thrust on one branch, while the same command landed on a sane branch at a neighboring rpm. No stabilizer fixes a model whose answer is not unique; the model itself is wrong for that regime. This is the failure the vortex-method literature predicts for separated stator flow[13] — arriving here not as a graceful loss of accuracy but as non-uniqueness, which is the more dangerous form because a single converged run gives no sign of it.

The default vane model therefore replaces the vane lattice with the closure a stator row actually obeys: annulus-by-annulus angular-momentum bookkeeping, the Euler turbomachine relation posed sector by sector.[42] Each vane strip owns one azimuthal sector of its radial annulus, carrying the mass flow the jet delivers,

$$\dot{m} = \rho\, u\, \frac{2\pi r}{N_s}\, \Delta r, \tag{6}$$

and the tangential force the strip may carry is exactly the angular momentum it removes from that flow. Blade-element aerodynamics supplies the same force from the section polar, evaluated at the *passage-mean* flow $(u,\ w - \Delta w/2)$, so the closure is one scalar equation per strip,

$$\dot{m}\, \Delta w = F_t(\Delta w), \tag{7}$$

monotone in Δw and solved by bisection on a bracket bounded by the local dynamic head. The consequences fall out rather than being imposed: every force is bounded by its sector's momentum flux, so the runaway channel does not exist; a dead-air strip carries nothing, so the jet-edge pathology closes itself; the undeflected cruciform cannot recover more torque than the jet delivers, so the swirl-budget factor, its root find, and its failure modes all become unnecessary on this path; and the solve is direct and single-valued — no iteration between strips, no branches, no warm starts. One derived normalization guards the seam to the slipstream: the azimuthal-mean reconstruction is not flux-consistent (it was measured delivering roughly three times the shaft torque to the vane plane), so the sampled swirl is scaled once, up front, to the jet's true angular-momentum flux — a computation, not an iterated feedback. The equivalent bound circulations still feed the rotor's azimuthal-mean feedback field, Eq. (4), so the two-way structure is unchanged.

What the closure gives up, stated plainly: no mutual induction between strips, no unsteadiness, and a passage-mean picture that assumes the vane row fills its annulus the way a stator row does. Those are the right trades at hover; in a brisk attached-incidence jet, where mutual strip induction is real signal rather than a runaway channel, the lattice closure remains available and validated.

VII. Results

Results are shown for a reference two-blade rotor with a snug shroud (4% tip clearance) and a four-vane cruciform at the duct exit (Fig. 2), at static (hover) conditions. All numbers are from the current build — Level-B wake, cascade closure, vane tips recessed off the duct wall (Sec. V).

Self-consistent wake pitch and the curved near wake. On the vane-less reference blade at hover the wake-pitch iteration converges in four outer passes, and the converged pitch is verified as a fixed point of the loading: rebuilding the lattice from the final $v_i(r)$ reproduces the trailing legs to 1.3×10^{-4}. The converged pitch varies more than threefold across the span (0.081–0.290 in axial pitch ratio), and a heavier-loaded blade converges to a 28% steeper mean helix (5.19 N against 3.77 N of thrust) — the load dependence and radial variation a fixed inflow constant cannot express. But the pitch was never where the hover optimism lived; the wake's *curvature* is. With straight self-consistent legs the reference blade returns a hover figure of merit of 1.22 — above the momentum-theory ideal of unity, an unphysical answer — and switching on the contracted helical filaments drops thrust and raises shaft torque together (7.65 N $\rightarrow$ 5.19 N at fixed rpm), bringing the figure of merit to 0.45, inside the physically admissible range. This is precisely the induced power a straight-leg wake cannot carry, and a collinear polyline is verified to reproduce the straight-leg solve identically.

Ducted rotor. At 6000 rpm under the curved wake the snug shroud lifts hover thrust from 6.76 N open to 7.05 N ducted (+4.3%), the duct itself carrying +0.26 N of bore-lip suction, with the coupled solve converged in 9 iterations and the blade circulation measurably shifted by the shroud's constriction. The augmentation is the interaction, not the presence of a ring: a loosely fitted ring earns almost nothing. Worth stating plainly: under the straight-leg wake this same comparison read 7.60 → 8.82 N (+16%) — an augmentation inflated by a baseline whose figure of merit exceeded unity. The curved wake deflates both numbers into the physical range, and the modest +4% is the honest potential-flow answer for this snug, short shroud at static conditions.

Vane system. Under the default cascade closure the coupled solve converges in 9 outer passes with no budget iteration at all, the vanes recovering 72% of the jet's angular-momentum flux on the reference case — about 70% across the rpm matrix (Table 1: $M_x^{vane}/Q = 0.0698/0.0998$ at 6000 rpm), at and never beyond the physical bound. The matrix coheres end to end: the rotor is perturbed by the vanes at the ∼1% level (5.31 N against 5.32 N alone at 6000 rpm), the duct carries a small positive thrust, and the undeflected vanes contribute *net thrust* through the pressure rise of swirl recovery (vane $F_x = -0.158$ N at 6000 rpm, thrust being $-F_x$) — a stator credit, not only a drag debit — with the system figure of merit at 0.48–0.50 across all four rpms, rising weakly with Reynolds number, under the ducted bound, with no tuned constants and no runaway anywhere. The single-vane control response (Table 2) is bounded, sign-correct, and asymmetric about the swirl-biased neutral — −55 mN of side force at $\delta = -8°$ against +24 mN at +8° — in the same swirl-rich jet that bifurcated the lattice, with deflection costing under half a percent of thrust at ±15°. The *vectored thrust* T_{vec} — the magnitude of the full force wrench, reported alongside the axial net thrust so a command trading axial thrust for side force is credited rather than penalized — nearly coincides with T_{net} for this cruciform, single-vane commands trading mostly into counter-torque at these deflections. The transpiration-coupled boundary-layer section model at 6000 rpm lands coherently beside the analytic polar ($T = 5.73$ N, FoM 0.50), settling on a stationary plateau that exits at the outer-pass cap — the boundary-layer rotor's inner-solve wobble, not the vane closure — and is reported unconverged rather than declared settled. The physical statement about jet-vane control at hover stands on either closure: authority near neutral comes from the vanes the swirl has *not* stalled, and grows as forward speed straightens the jet.

Table 1. Hover performance across rpm: undeflected cruciform, cascade closure, analytic section polar. Net thrust and its rotor, duct, and vane shares in N (thrust = $-F_x$), shaft torque Q in N·m, power P in W; *passes* is the outer Gauss–Seidel count.

rpm	T_{net}	T_{rotor}	T_{duct}	vane F_x	Q	P	FoM	passes
3000	1.367	1.326	0.007	−0.040	0.0254	7.97	0.482	9
6000	5.473	5.314	0.029	−0.158	0.0998	62.7	0.491	9
9000	12.318	11.966	0.067	−0.352	0.2226	209.8	0.495	9
12000	21.903	21.281	0.121	−0.623	0.3935	494.5	0.498	9

Table 2. Single-vane control response at 6000 rpm, cascade closure: one vane deflected by δ, the other three at neutral. Forces in N, moments in N·m.

δ, deg	T_{net}	T_{vec}	F_z	M_y	M_x^{vane}	passes
−15	5.448	5.449	−0.124	+0.0069	0.0571	11
−8	5.465	5.465	−0.055	+0.0030	0.0641	10
0	5.473	5.473	0	0	0.0698	9
+8	5.473	5.473	+0.024	−0.0012	0.0725	9
+15	5.451	5.451	+0.023	−0.0011	0.0725	9

Consistency with swirl-recovery-vane measurements. No external validation is claimed here — that anchor is being assembled for the journal version — but the two headline behaviours of the coupled solve are the ones the experimental swirl-recovery-vane literature reports. Vanes standing in a propeller's slipstream consume no shaft power, so any streamwise force they produce is a direct credit to propulsive efficiency; wind-tunnel campaigns measure a thrust increase of order 2.6% at the design point, together

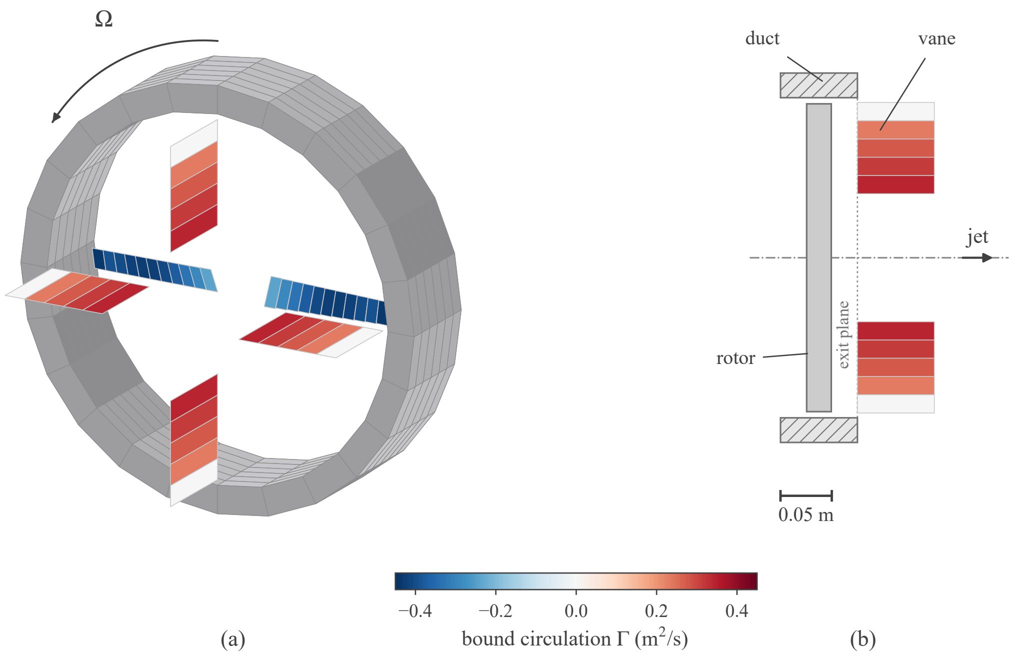

Figure 2. The reference configuration, drawn from the solved lattice (6000 rpm, cascade closure, neutral vanes): (a) aft quarter view, duct shroud source panels in gray, blades and vanes colored by solved bound circulation — the two populations carry opposite signs, the cruciform recovering the rotor's swirl; (b) meridional section, the vane chords running downstream from their exit-plane hinge lines, the rotor disk edge-on, and the in-plane vane pair at true profile.

with a roughly 50% reduction of the slipstream's swirl kinetic energy at medium thrust settings.[40,43] The present cruciform behaves the same way in kind — a net thrust credit from the undeflected vanes (Table 1) and about 70% of the jet's angular-momentum flux recovered as counter-torque — with the larger recovery fraction consistent with the far higher swirl of a static ducted fan than of a cruise propeller. The agreement is qualitative and is reported as such.

Verification. Every claim above is pinned by the automated test suite: exact cruciform symmetry undeflected, counter-torque opposing the rotor's, the budget bound $M_x^{vane} \leq Q$ at convergence on both closures, opposite commands pulling the wrench opposite ways, drag cost for commanding against the swirl, dead-air strips carrying nothing under the cascade closure, and the survival of the control response under the two-way coupling.

VIII. Conclusion

A ducted fan with exit vanes can be resolved as one coupled potential-flow system — rotating-frame vortex lattice for the blades, source rings for the duct, vane loads read from a momentum-theory slipstream reconstructed from the solved loading — with the swirl exchange between the frames closed by an angular-momentum budget. Two closures implement it: a lifting-surface vane lattice whose recovered torque must be constrained by solving the budget as a bracketed root problem with the rotor frozen (a feedback law was observed to fail in both directions), and the default cascade momentum closure, which bounds every strip's force by its sector's momentum flux by construction and needs no budget iteration at all — the choice the vane lattice's post-stall bistability at hover forces. The budget is not a refinement but a correction of leading order: the unconstrained scheme overpredicted the recovered vane torque several-fold on the reference configuration, while the coupled scheme converges to the physically bounded state in a handful of outer passes and seconds of wall time. The method gives a ducted-fan VTOL designer the quantities the configuration loop actually needs at hover — net thrust with the vanes' drag debit, control moments about the swirl-biased neutral, and the recovered counter-torque — from a parametric geometry description, at interactive speed.

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